Advanced Programming Practice · 21CSC203P

Unit 4 · Session 3 · SLO 2 — Solved Worksheet

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Unit â€“ 4 : Session â€“ 3 : SLO â€“ 2 | Lambda Expressions and Anonymous Functions

I. Programming Task â€” Custom Higher-Order Function

```
def transform(items, func):
"""Apply func to every element of items and return the new list."""
return [func(x) for x in items]

# usage
nums = [1, 2, 3, 4]
print(transform(nums, lambda x: x * x)) # [1, 4, 9, 16]
print(transform(nums, lambda x: x + 10)) # [11, 12, 13, 14]
```

II. Explanation

- transform() is higher-order because it accepts another function (func) as a parameter.
- The lambda expressions are anonymous functions defined inline â€” no def or name needed.
- This pattern generalizes map(): any transformation can be plugged in without changing transform() itself.